

The transshipment problem

Total unimodularity

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Optimization: principles and algorithms

Total unimodularity

The incidence matrix defining the constraints of a transshipment problem has a special structure

It also has a great property called “total unimodularity”.

It is great because it guarantees to generate an integer solution, if the data is integer.

In this video, we introduce the definition of this property.

Transshipment problem in standard form

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

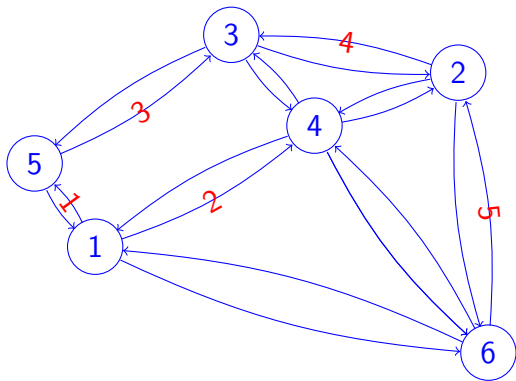
$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N},$$
$$x_{ij} \geq 0, \forall (i,j) \in \mathcal{A}.$$

subject to

$$\min_{x \in \mathbb{R}^n} c^T x$$

$$Ax = s,$$
$$x \geq 0.$$

Incidence matrix



	1	2	3	4	5	...
1	1	1	0	0	0	
2	0	0	0	1	-1	
3	0	0	-1	-1	0	
4	0	-1	0	0	0	
5	-1	0	1	0	0	
6	0	0	0	0	1	

Optimal solution

$$x_B^* = B^{-1}s.$$

Cramer's rule

$$x_B^* = \frac{1}{\det(B)} C(B)^T s$$

Each entry of the matrix is 0, 1 or -1.

Each entry of $C(B)$ is an integer.

If s is integer, $C(B)^T s$ is integer.

If $\det(B)$ is 1 or -1, x_b^ is integer.*

Total unimodularity

Definition

The matrix $A \in \mathbb{Z}^{m \times n}$ is **totally unimodular** if the determinant of each square submatrix of A is

$$0, -1 \text{ or } +1.$$

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The matrix $A \in \mathbb{Z}^{m \times n}$ is **totally unimodular** if the determinant of each square submatrix of A is

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In particular,

every entry of the matrix is 0, -1 or $+1$.

Example

Each square submatrix of size 1, that is each element of the matrix, is 0, -1 or +1. Each square submatrix of size 2 is unimodular:

$$A = \begin{pmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \end{pmatrix}.$$

$$\det \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = 1,$$

$$\det \begin{pmatrix} -1 & 0 \\ 1 & 1 \end{pmatrix} = -1,$$

$$\det \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} = 1.$$

Integrality of the basic solutions

Consider the polyhedron represented in standard form

$$\{x \in \mathbb{R}^n \mid Ax = b, x \geq 0\},$$

where $A \in \mathbb{Z}^{m \times n}$ and $b \in \mathbb{Z}^m$.

If A is totally unimodular, then every basic solution is integer.

Proof

$$x_B = B^{-1}b. \quad x_B = \frac{1}{\det(B)} C(B)^T b,$$

Summary

We have introduced the concept of “totally unimodular” matrix

Each square submatrix has determinant 0, -1 or +1.

In this case, each basic solution of a linear optimization problem with A totally unimodular and b integer is integer.

Next, we show that the incidence matrix of the transshipment problem is totally unimodular.